Chapter 1: Introduction

Key Purpose:
This chapter introduces the subject of economics, emphasizing that every field, including economics, has its own language and way of thinking.
Main Ideas:
- Economics uses specialized terms like *demand*, *supply*, and *market*.
- Just like in mathematics or science, learning to think like an economist takes time and practice.
- The book aims to help students understand how economists think using theory, examples, and practical problems.

Next Step:
Before diving into the meaning of economics, it's important to first understand what an economy is and why studying economics is important.
What is an Economy?
- An **economy** is a system that includes institutions (like factories, shops, railways, etc.) which help people earn a living and produce goods/services.
- It uses **available resources** to fulfil people's wants and needs.
- Example: The Indian economy includes agriculture, industry, transport, banking, etc.

Vital Processes of an Economy
Every economy performs **three essential activities**:
1. **Production** – Making goods and services
2. **Consumption** – Using goods and services
3. **Investment or Capital Formation** – Creating resources for future production

Why Study Economics?
- Economics helps us understand how various activities in the economy function.
- The **main reason to study economics is scarcity** — our wants are unlimited, but resources are limited.
- Economics deals with **making choices** and **allocating scarce resources** efficiently.

Scarcity
- **Scarcity** occurs when **demand exceeds supply**.
- It is **universal** — every person and economy face it.
- Leads to the need for **economizing**, i.e., **optimally using limited resources**.
- Resources are not only scarce but also have **alternative uses**, meaning they can be used in multiple ways.

Economic Problem
What is an Economic Problem?
- Human wants are **unlimited**, **differ in priorities**, but resources to satisfy them are **limited** & Have **alternative uses.**
- This mismatch creates the **economic problem**, which is basically a **problem of choice**.
- People must decide **what to produce, how to produce, and for whom to produce**, because resources have **alternative uses**.

Economic Problem = Choice + Scarcity + Alternative Uses

Main Reasons for Economic Problem
1. **Scarcity of Resources**
   - Resources like land, labour, and capital are limited.
   - Economy can't produce everything people want.
2. **Unlimited Human Wants**
   - Human wants never end; new ones keep arising.
   - Wants also vary in **urgency and importance**, forcing people to prioritize.
3. **Alternative Uses of Resources**
   - Resources can be used in many ways.
   - Choosing one use means giving up others (e.g., petrol in vehicles vs machines).

Meaning of Economics
- **Economics** is a **social science** that studies how societies **use limited resources** with **alternative uses** to produce and distribute goods and services.
- It is called a **social science** because it systematically studies **human behaviour**, though it's not as exact as physical sciences.
- **Core focus**: Economics is about **making choices in the presence of scarcity**, ensuring optimal use of resources to satisfy human wants.

Positive and Normative Economics
Positive Economics
- Describes **economic facts and realities** ("what is", "what was", "what will be").
- Deals with actual issues like rising prices or population growth.
- **Does not make judgments**—just states facts that can be **tested or verified**.
- Example: "India is overpopulated."

Normative Economics
- Involves **opinions and value judgments** ("what ought to be").
- Suggests how economic issues **should** be solved.
- Example: "India should reduce its population."

Examples	1. Prices are rising2. Income inequality exists	1. Prices **should be controlled**2. Inequality **should be reduced**

Microeconomics and Macroeconomics
1. **Microeconomics (Price Theory)**
   - **Micro = Individuals ('I')** → Price, demand, supply.
   - Term derived from Greek **'mikros' = small**
   - Studies **individual units**: consumers, firms, small groups
   - Founded by **Adam Smith**
   - Important for understanding how small units' **function within an economy**
   - Studies behaviour and decision making of **individual units** (consumers, firms).
   - Tools: **Demand and Supply**.
   - Objective: **Price determination** of goods/factors.
   - Limited aggregation (market demand = sum of individual demands).
   - Assumes **macro variables constant**.
   - Examples: Individual income, individual output.

   Macroeconomics (Income & Employment Theory.)
   - **Macro = Aggregates ('A')** → Income, employment, national output.

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- **Origin**: Derived from Greek word *"makros" = large*.
- **Focus Areas**: Problems like **inflation, unemployment, poverty**.
- **Definition**: Part of economic theory that studies the behaviour of **aggregates** of the economy as a whole.
- **Examples**: National income, aggregate output, aggregate consumption.
- **Main Tools**: **Aggregate Demand** and **Aggregate Supply**.
- Studies behaviour of the **economy as a whole**.
- Objective: **Income & employment determination**.
- Highest aggregation (entire economy).
- Assumes **micro variables constant**.
- Examples: National income, national output

Interdependence: - Micro and Macro **cannot be separated**; both are interdependent

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Central Problems of an Economy: - Because resources are **scarce** and wants are **unlimited**, every economy faces **3 central problems** (Allocation of Resources):

1. What to Produce?
   - Decide **which goods & services** to produce (consumer vs. capital goods, civil vs. war goods).
   - Decide **how much** of each good to produce.
   - Example: More sugar → less of other goods; more war goods → less civil goods.
   - **Guiding Principle:** Allocate resources to give **maximum satisfaction**.
2. How to Produce?
   - Decide **technique of production**:
     - **Labour-intensive** (more labour, less capital).
     - **Capital-intensive** (more capital, less labour).
   - Choice depends on availability of resources & cost.
   - Example: India (abundant labour) → LIT; USA/UK (abundant capital) → CIT.
   - **Guiding Principle:** Use factors of production so that **maximum output at minimum cost** is achieved.
3. For Whom to Produce?
   - Decide **who will consume goods** → based on **income distribution**.
   - Since wants > resources, goods go to those with **paying capacity**.
   - Output distributed as:
     - Rent → Land
     - Wages → Labour
     - Interest → Capital
     - Profit → Enterprise
   - **Two Aspects:**
     1. **Personal distribution** → Income distribution among people.
     2. **Functional distribution** → Share of factors of production in national product.
   - **Guiding Principle:** Ensure **urgent wants** of each productive factor are fulfilled.

📌 **In Short:**

1. **What to Produce** – Selection of goods/services and their quantities.
2. **How to Produce** – Choice of technique: Labour Intensive or Capital Intensive.
3. **For Whom to Produce** – Distribution of goods and services among individuals.

Opportunity Cost

- Resources are scarce → choices must be made.
- **Definition:** Opportunity cost = cost of next best alternative foregone.
- Example 1: Salary choices → If bank job pays ₹70,000 but next best option is journalist at ₹65,000, then opportunity cost = ₹65,000.
- Example 2: Buying MacBook instead of QLED TV → foregone satisfaction of TV is opportunity cost.
- **Key Point:** We always face trade-offs due to limited resources.

Production Possibility Frontier (PPF)

- PPF shows the maximum **possible combinations of two goods** that can be produced with **given resources & technology**.
- Example often used: **Guns & Butter**.
- Graphical representation of **Production Possibility Schedule**. (PPS is the table showing different combinations)
- **Represents choice & scarcity** → since resources are limited, producing more of one good means sacrificing some of the other.
- Reflects economic choices and trade-offs.

Synonyms of PPF

- **Production Possibility Curve (PPC), Production Possibility Boundary, Production Possibility Frontier**
- **Transformation Curve, Transformation Boundary, Transformation Frontier**

Assumptions of PPF

1. Total resources are **fixed** but transferable between uses.
2. Only **two goods** are produced.
3. Resources are **fully and efficiently utilised**.
4. Resources are **not equally efficient** → shifting resources lowers productivity.
5. Technology is **constant**.

Example (Hypothetical Schedule)

Possibilities	Guns	Butter	MRT (ΔG/ΔB)
A	21	0	—
B	20	1	1G : 1B
C	18	2	2G : 1B
D	15	3	3G : 1B
E	11	4	4G : 1B
F	6	5	5G : 1B
G	0	6	6G : 1B

✅ **Key Point:** PPF highlights **scarcity, choice, and opportunity cost** in an economy.

PPF, MOC & MRT

- **Marginal Opportunity Cost (MOC):**
  - Refers to the amount of one good sacrificed to gain an extra unit of another.
  - **MOC always increases** because resources are not equally efficient in producing all goods.
- **Why MOC increases:**
  - As resources shift from one good to another, productivity falls → more of one good must be sacrificed.

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- **Marginal Rate of Transformation (MRT):**
  - Ratio of sacrifice of one good to gain another.
  - MRT is measured by the slope of the PPF.
  - On a concave PPF, slope increases as we move downward → MRT rises
  - Formula: MRT=Δ Guns /Δ Butter
  - Example: To produce 1B, 2G must be sacrificed → MRT = 2G : 1B.

Properties of PPF

1. **Slopes Downwards** – To produce more of one good, some of the other must be sacrificed (inverse relationship).
2. **Concave Shape** – Due to *increasing marginal rate of transformation (MRT)*. Resources are not equally efficient, so more of one good requires sacrificing increasing amounts of the other.

Economy's Operation on PPF

- Economy operates **on PPF** if resources are fully & efficiently used.
- Economy operates **inside PPF** if resources are underutilized.
- Economy **cannot operate outside PPF**, as it exceeds productive capacity.

Attainable vs. Unattainable Combinations

- **Attainable:** Points on PPF (A, B, C, D) or inside PPF (E).
- **Unattainable:** Points outside PPF (like F).
- **Optimum Utilisation:** Economy operates on the PPF curve itself.

1. **Inefficient Utilisation of Resources**
   - If resources are wasted or underutilised, production occurs *inside* the PPF (e.g., Point E).
2. **Unattainable Combinations**
   - Economy *cannot* produce beyond PPF (e.g., Point F) due to limited resources.
3. **PPF as a Straight Line**
   - Possible only if MRT is constant (resources equally efficient in all goods).
   - Rare in real life.
4. **PPF as Convex to Origin**
   - Possible only if MRT is decreasing (less sacrifice needed).
   - In practice, this doesn't happen.
5. **Conclusion**
   - Since MRT always increases, **PPF is always concave-shaped.**

- **Transformation Curve:** PPF is also called Transformation Curve as it shows how resources are shifted between goods (e.g., guns ↔ butter).

1. **Shift in PPF**
   - **Rightward Shift:** Occurs due to growth of resources or technological **advancement for both goods** → economy can produce more of both goods.
     - Causes: Discovery of resources, foreign capital inflow, more labour, skill development, better hygiene (e.g., Swachh Bharat Mission).
   - **Leftward Shift:** Happens due to fall in resources or technological degradation (e.g., earthquake) → economy produces less of both goods.
2. **Rotation of PPF**
   - Happens when productive capacity changes for **only one good**.
   - Example: If resources/technology increase for butter (X-axis good), PPF rotates outward for butter only.

Rotation of PPF

1. **Rotation on X-axis (butter):**
   - Technological improvement/resources ↑ → PPF rotates right (AB → AC).
   - Technological degradation/resources ↓ → PPF rotates left (AB → AD).
2. **Rotation on Y-axis (guns):**
   - Technological improvement/resources ↑ → PPF rotates right (AB → CB).
   - Technological degradation/resources ↓ → PPF rotates left (AB → DB).

Overview of PPF

- **Slopes downwards** → producing more of one good requires sacrificing another.
- **Concave shape** → due to increasing MRT (opportunity cost rises).
- **Transformation** → shows reallocation of resources from one good to another.
- **Possibilities** → shows maximum production combinations.
- **On the curve (A, B, C):** full & efficient use of resources.
- **Inside curve (D):** underutilisation of resources.
- **Outside curve (E):** unattainable with current resources.
- **Outward shift (PP → P1P1):** economy can produce more of both goods (growth).

✅ **Key Takeaways:**

- Opportunity cost = best alternative sacrificed.
- PPF illustrates trade-offs and efficiency.
- MOC increases as more of one good is produced (concave PPF).

✅ **Key Idea**:

- **Decreasing MRT → Convex PPF**
- **Increasing MRT → Concave PPF**

Revision of Key Points (Summary)

- **Economy**: System providing people with means to earn a living.
- **Scarcity**: Limited supply vs. unlimited demand.
- **Economising resources**: Optimum use of resources.
- **Economic problem**: Choice between unlimited wants and limited resources.
- **Economics**: Social science of using limited resources for production & distribution.

- **Opportunity Cost**: Next best alternative forgone.
- **Production Possibility Frontier (PPF)**:
  - Downward sloping (trade-off between goods).
  - Concave shape (increasing MRT).
  - Points inside = inefficient, on = efficient, outside = impossible.
  - **Rotation** → change in tech/resources of one good.
  - **Shift** → change in resources/tech of both goods (rightward = growth, leftward = decline).
- **Synonyms**:

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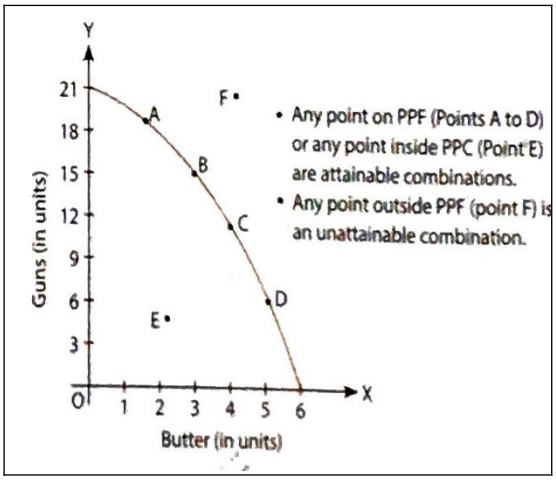
- Positive Economics = Positive Science
- Normative Economics = Normative Science
- Microeconomics = Price Theory
- Macroeconomics = Income & Employment Theory
- PPF = PPC, Production Boundary, Transformation Curve/Frontier



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